Vasu Ganesa

EDUCATION

American High School Class of 2027
GPA: 3.97 unweighted
Relevant Courses:

- Completed Multivariable Calculus at Mission College with an A grade, a sophomore-level college course beyond AP Calculus BC.
- Currently taking Linear Algebra at West Valley College, a sophomore-level college course beyond AP Calculus BC.

Relevant Clubs: Math Club President
Portfolio: https://github.com/einstefer

PROFESSIONAL EXPERIENCE

Full Stack Machine Learning Engineer | MIT CSAIL **June 2026 - Present**

- Developed eight software features across Mantis AI and MantisGPT, an AI-powered educational data exploration tool developed within the MIT Computer Science & Artificial Intelligence Laboratory Kellis Lab.
- Implemented multi-file CSV/XLSX ingestion, visualization enhancements, UI/UX improvements, and performance optimizations for educational data analysis.

AI & Data Analysis Intern | Educational Technology Gnosis Learning Inc. **April 2026 – Present**

- Contributed to the development of an AI-powered education platform that delivers personalized AP and K–12 learning through intelligent eBooks, adaptive study tools, and interactive AI tutoring.
- Developed software to analyze learner engagement, content effectiveness, and platform performance to evaluate educational outcomes.

Full Stack Web Development Intern | Lift and Shift **June 2025 – Present**

- Developed the Lift and Shift Foundation games website and implemented five interactive web-based games using Python, JavaScript, HTML, and CSS.
- Developed and launched a cross-platform equine veterinary management application on the Google Play Store.

Math Tutor and Student CEO | Micro Lessons Academy (Non-Profit) **August 2023 – Present**

- Advanced from volunteer tutor to student CEO and oversaw outreach, curriculum planning, and recruitment for a nonprofit focused on accessible math education.
- Expanded outreach to California, Oregon, and Washington by cold-calling approximately 27 libraries and cold-emailing approximately 235 libraries to promote summer math classes.
- Created the nonprofit's first-ever Summer Math Camp, reaching approximately 46 students across 3 states with structured math enrichment.
- Led the organization's weekly AMC 8 preparation program for 50 regular attendees, teaching competitive math concepts and guiding interactive problem-solving sessions.

Math Club President | American High School **August 2023 – Present**

- Created weekly lesson plans, slides, and practice materials for 50+ students, covering problem-solving strategies and advanced competition math topics.
- Managed high school math contest preparation and logistics for competitions such as BMT and SMT consisting of 100 attendees, including registration support, team coordination, and student preparation.
- Organized club events and outreach initiatives to promote math engagement, increase student participation, and strengthen the school's competitive math community.

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VOLUNTEER WORK

Math Tutor | Hayward Library **August 2024 – Present**

- Tutored middle school students in math on a weekly basis as part of a pro-bono community program.
- Supported 300+ students through recurring local library tutoring sessions focused on homework help, foundational math skills, and problem-solving strategies.
- Helped create an accessible learning environment for students seeking free academic support in mathematics.

MathCounts Volunteer Coach | Thornton Junior High School **January 2023 – Present**

- Coached and mentored middle school students through a multi-stage MathCounts preparation program, supporting a cohort selected from 275 initial sign-ups and 261 screening test participants.
- Helped lead 40 qualified students through several weeks of in-person and online coaching, developing practice materials, teaching problem-solving strategies, and preparing students for school and chapter-level competitions.
- Supported 14 students who advanced to the Fremont Chapter Competition, where Thornton students earned 3 top-10 individual placements out of 70+ competitors.
- Coached Thornton's 4-person team to a 2nd-place Fremont Chapter finish and qualification for the MathCounts State Competition for the first time in Thornton Junior High School's history.

RESEARCH EXPERIENCE

Research Assistant to Dr. Xiucai Ding | UC Davis Department of Statistics **April 2026 – Present**

- Analyzed high-dimensional biomedical and EEG datasets for a brain-computer interface backed by UC Davis medical team and supported by a National Science Foundation grant.
- Applied statistical learning, dimensionality reduction, and deep learning techniques to evaluate 11 machine learning methods for P300 EEG classification across 110 subjects.
- Authored a sole-author P300 EEG research paper accepted to IEEE ICIIBMS 2026.

Research Assistant to Dr. Padmanabhan Seshaiyer | George Mason University **April 2026 – Present**

- Conducted research on mathematical modeling approaches related to UN Sustainable Development Goal 4: Quality Education, with a focus on mathematics education.
- Explored how applied mathematical models can support improved access, equity, and learning outcomes in math education.

LLM and Math Research with Lars Holdijk | Oxford PhD Student **December 2024 – Present**

- Investigated how 5 prompting strategies affected the mathematical reasoning performance of 3 Large Language Models (LLMs) across 60 American Invitational Mathematics Examination (AIME) problems.
- Designed Python evaluation pipelines and analyzed prompting performance using Cochran's Q and McNemar's statistical tests with multiple-comparison correction.
- Published in the Journal of Emerging Investigators (https://emerginginvestigators.org/articles/25-137)

AWARDS

- United States of America Junior Mathematical Olympiad Qualifier (Top ~250 nationwide out of 300,000+ AMC participants; first and only qualifier in American High School's history)
- AMC 10 A & B Distinguished Honor Roll (Top 1%)
- AMC 12 A & B Honor Roll (Top 5%)
- 3× AIME Qualifier (Top ~6% nationally on AMC 10)
- PSAT - 1510/1520 (750 E, 760 M)
- SAT - 1540/1600 (740 E, 800 M)